



MANIPAL INSTITUTE OF TECHNOLOGY

BENGALURU

(A constituent unit of MAHE, Manipal)

B.Tech Fourth Year

Course Handout

Session (Semester):	21 st July, 2025 to 12 th November, 2025 (VII)			
Branch:	Computer Science and Engineering (CSE-CORE)			
Class:	Theory Course			
Course Name (Code):	Machine Learning-IV (CSE_4032)			
Contact Hours/Week:	L	T	P	C
	3	0	0	3
Course Coordinator :	Dr. Manaswita Saikia			
Course Instructor:	Dr. Manaswita Saikia			

A. Introduction:

This course provides a fundamental and applicable understanding of machine learning principles and methods, starting with supervised learning and progressing to unsupervised models and ensemble techniques. Students will learn about every aspect of the machine learning design cycle, including issue conceptualization and data preparation and continuing with assessment of models and deployment. Bayesian logic, regression and classification models, support vector machines, and various clustering methods are among the key topics explored. Fundamental problems such as overfitting, the bias-variance tradeoff, and model generalization are addressed, in addition to metrics for performance and optimization techniques. The course also covers decision trees and ensemble techniques such as bagging and boosting to enhance model accuracy. The emphasis is on both conceptual clarity and present application, preparing students to develop and assess intelligent systems in real-world settings.

B. Objectives:

The main objective of the course is as follows:

- To understand the theoretical underpinnings of Machine Learning
- To have basic knowledge about the key algorithms and theory that forms the foundation of machine learning.
- To have a practical knowledge of machine learning algorithms and methods.

C. Assessment Rubrics:

Assessment No.	Assessment Type	Marks	Weightage
Assessment 1	Surprise Test (Offline)	5	5
Assessment 2	Quiz (LMS)	5	5
Assessment 3	Activity Based Assessment	5	5
Assessment 4	Assignment (LMS)	5	5
Assessment 5	Mid Term Exam (EPad)	30	30
Total Internal Marks			50
External	End Term Exam (EPad)	50	50
Total Marks			100
A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking up the End Semester examination, but this attendance limit is not mandatory for the online classes as guided by AICTE. The allowance of 15% includes all types of leaves including medical leaves			

D. Syllabus:

MODULE-1 INTRODUCTION and SUPERVISED LEARNING BASICS

Towards Intelligent Machines, Well-Posed Machine Learning Problems, Examples of Applications in Diverse Fields, Forms of Learning: Supervised Learning, Unsupervised Learning, Reinforcement Learning.

Learning from Observations, Bias and Variance, Computational Learning Theory, Occam's Razor Principle and Overfitting Avoidance, Metrics for Assessing Regression Accuracy, Metrics for Assessing Classification Accuracy, An Overview of the Design Cycle and Issues in Machine Learning

MODULE-2 STATISTICAL LEARNING Bayesian Reasoning: Bayes Theorem, Naïve Bayes Classifier, Bayesian Belief Networks, k-Nearest Neighbour Classifier, Discriminant Functions and Regression Functions, Linear Regression with Least Square Error Criterion, Logistic Regression for Classification Tasks.

Parametric Methods: Introduction, Maximum Likelihood Estimation, The Bayes Estimator.

MODULE-3 SUPPORT VECTOR MACHINES

Introduction, Linear Discriminant Functions for Binary Classification, Perceptron Algorithm, Linear Maximal Margin Classifier for Linearly Separable Data, Linear Soft Margin Classifier for Overlapping Classes, Regression by Support Vector Machines: Linear Regression, Nonlinear Regression

MODULE-4 DATA CLUSTERING

Unsupervised Learning, Engineering the Data, Overview of Basic Clustering Methods, K-Means Clustering, Expectation-Maximization Algorithm and Gaussian Mixtures Clustering, Hierarchical Clustering

MODULE-5 DECISION TREES

Introduction, Example of a Classification Decision Tree, Measure of Impurity for Evaluating Splits in Decision Trees, ID3, C4.5, and CART Decision Trees, Pruning the Tree, Strengths and Weaknesses of Decision Tree Approach

MODULE-6 COMBINING MULTIPLE LEARNERS

Rationale, Generating Diverse Learners, Voting, Error-Correcting Output Codes, Bagging, Boosting.

E. Text Books:

1. M. Gopal, *Applied Machine Learning*, McGraw Hill Education, 2018.
2. Ethem Alpaydin, *Introduction to Machine Learning*, 2nd edition, MIT Press. 2010.

F. Reference Books:

1. Peter Harrington, *Machine Learning in Action*, Manning Publications, 2012.
2. Andreas C. Müller & Sarah Guido, *Introduction to Machine Learning with Python*, O'Reilly Media Inc., 2017.
3. Tom M Mitchell, *Machine Learning*, McGraw Hill, 2017

G. Course Outcomes:

- CO.1** To explain basic concepts of machine learning and supervised learning.
CO.2 To apply statistical methods for classification and inference.
CO.3 To use support vector machines for classification and regression.
CO.4 To apply unsupervised learning methods for clustering.
CO.5 To use decision trees for classification.
CO.6 To apply ensemble learning techniques for improved classification.

H. Program Outcomes:

- [PO.1]. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- [PO.2]. Problem analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- [PO.3]. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- [PO.4].** Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- [PO.5].** Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

[PO.6]. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.

[PO.7]. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

[PO.8]. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices.

[PO.9]. Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

[PO.10]. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

[PO.11]. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

[PO.12]. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

I. Program Specific Course Outcomes:

[PSO1]. Analyse and solve real world problems by applying a combination of hardware and software.

[PSO2]. Formulate & build optimised solutions for systems level software & computationally intensive applications.

[PSO3]. Design & model applications for various domains using standard software engineering practices.

[PSO4]. Design & develop solutions for distributed processing & communication.

J. Lecture Plan:

Lect ures		Topics	Mode of Delivery	Corresp onding CO
L1	INTRODUCTION and SUPERVISED LEARNING BASICS	TOWARDS INTELLIGENT MACHINES: Needs, Motivations, and Rationale, Soft Computing/ Machine Learning, WELL- POSED MACHINE LEARNING PROBLEMS: EXAMPLES OF APPLICATIONS IN DIVERSE FIELDS, FORMS OF LEARNING: Supervised/Directed Learning, Unsupervised/ Undirected Learning, Reinforcement Learning, Learning Based on Natural Processes: Evolution, Swarming, and Immune Systems	Lecture	CO.1

L2		LEARNING FROM OBSERVATIONS: BIAS AND VARIANCE: WHY LEARNING WORKS: COMPUTATIONAL LEARNING THEORY	Lecture	CO.1
L3		Bias & Variance: Empirical risk minimization, Bias-Variance decomposition, Underfitting vs Overfitting, Overfitting Avoidance & Occam's Razor Principle: Regularization (L1/L2), early stopping, Simplicity in models – Occam's Razor, Role in hypothesis selection.	Lecture	CO.1
L4		Evaluation Metrics for Supervised Learning: Regression Metrics- MAE, MSE, RMSE, R ² Score.	Lecture	CO.1
L5		Evaluation Metrics for Supervised Learning: Classification Metrics-Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix	Lecture	CO.1
L6		An Overview of the Design Cycle and Issues in Machine Learning	Lecture	CO.1
L7	STATISTICAL LEARNING	BAYESIAN REASONING: A PROBABILISTIC APPROACH TO INFERENCE: Bayes Theorem, General Bayes Theorem, Naive Bayes Classifier	Lecture	CO.2
L8		Bayesian Belief Networks, Training Bayesian Belief Networks	Lecture	CO.2
L9		Eager Vs Lazy Learner, Distance metrics: Euclidean, Manhattan, k-NEAREST NEIGHBOR (k-NN) CLASSIFIER: Instance-based learning, , Curse of dimensionality.	Lecture	CO.2
L10		DISCRIMINANT FUNCTIONS AND REGRESSION FUNCTIONS: Classification and Discriminant Functions, Numeric Prediction	Lecture	CO.2
L11		Regression Functions, Practical Hypothesis Functions	Lecture	CO.2
L12		LINEAR REGRESSION WITH LEAST SQUARE ERROR CRITERION: Minimal Sum-of-Error-Squares and the Pseudoinverse, Gradient Descent Optimization Schemes	Lecture	CO.2
L13		Logistic Regression for Binary Classification: Sigmoid function, Maximum Likelihood Estimation of Logistic Regression Models	Lecture	CO.2
L14		Parametric Methods: Introduction, Maximum Likelihood Estimation, Bernoulli Density, Multinomial Density, Gaussian (Normal) Density, The Bayes' Estimator	Lecture	CO.2
L15		SUPPORT VECTOR MACHINES: INTRODUCTION, LINEAR DISCRIMINANT FUNCTIONS FOR BINARY CLASSIFICATION,	Lecture	CO.3
L16		PERCEPTRON ALGORITHM,	Lecture	CO.3

L17		LINEAR MAXIMAL MARGIN CLASSIFIER FOR LINEARLY SEPARABLE DATA,	Lecture	CO.3
L18		LINEAR SOFT MARGIN CLASSIFIER FOR OVERLAPPING CLASSES,	Lecture	CO.3
L19		REGRESSION BY SUPPORT VECTOR MACHINES: Linear Regression	Lecture	CO.3
L20		REGRESSION BY SUPPORT VECTOR MACHINES: Nonlinear Regression	Lecture	CO.3
L21	DATA CLUSTERING	UNSUPERVISED LEARNING: Clustering, ENGINEERING THE DATA: Exploratory Data Analysis: Learning about What is in the Data	Lecture	CO.4
L22		Cluster Analysis: Finding Similarities in the Data, Representing Clustering Result, Measuring Clustering Quality, Cluster Validity, Cluster Interpretability	Lecture	CO.4
L23		Cluster Stability: Applications of Cluster Analysis, Clustering for Data Exploration, Clustering as a Standalone Tool	Lecture	CO.4
L24		Data Transformations: Enhancing the Information Content of the Data, Data Cleansing, Derived Attributes, Discretizing Numeric Attributes, Attribute Reduction	Lecture	CO.4
L25		OVERVIEW OF BASIC CLUSTERING METHODS: Partitional Clustering, K-Means Algorithm, Fuzzy K-Means Clustering, Probabilistic Clusters, Hierarchical Clustering, Spectral Clustering, Clustering using Self-Organizing Maps	Lecture	CO.4
L26		K-MEANS CLUSTERING	Lecture	CO.4
L27		EXPECTATION-MAXIMIZATION (EM) ALGORITHM AND GAUSSIAN MIXTURES: EM Algorithm, Gaussian Mixture Models, , Hierarchical Clustering	Lecture	CO.4
L28	DECISION TREES	Decision Trees: INTRODUCTION, EXAMPLE OF A CLASSIFICATION DECISION TREE, MEASURES OF IMPURITY FOR EVALUATING SPLITS IN DECISION TREES: Information Gain/Entropy reduction, Gain Ratio, Gini Index.	Lecture	CO.5
L29		ID3, C4.5, AND CART DECISION TREES: The ID3 Decision Tree	Lecture	CO.5
L30		ID3, C4.5, AND CART DECISION TREES: The C4.5 Decision Tree	Lecture	CO.5
L31		ID3, C4.5, AND CART DECISION TREES: The CART Decision Tree,	Lecture	CO.5
L32		PRUNING THE TREE, STRENGTHS AND WEAKNESSES OF DECISION-TREE	Lecture	CO.5

		APPROACH: Feature Extraction using Trees, Rule Extraction from Trees, Regression Trees, Multivariate Trees,		
L33		Decision Tree is a Nonlinear Classifier, Decision Tree is a Non-parametric Model, Handling Missing Values and Outliers, When are Decision Trees Appropriate?	Lecture	CO.5
L34	COMBINING MULTIPLE LEARNERS	COMBINING MULTIPLE LEARNERS: Combining Multiple Learners, Rationale, Generating Diverse Learners, Voting	Lecture	CO.6
L35		Error-Correcting Output Codes	Lecture	CO.6
L36		Bagging, Boosting	Lecture	CO.6

K. Course Articulation Matrix: (Mapping of COs with POs):

		PO 1	P O 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	PS O4
CO 1	Explain basic concepts of ML & supervised learning	3	2	1	1	1							1		1	1	
CO 2	Apply statistical methods for classification & inference	3	2	1	1	1							1		1	1	
CO 3	Use SVM for classification and regression	3	2	1	1	1							1		2	2	
CO 4	Apply unsupervised learning methods for clustering	3	2	1	1	2							1		2	2	
CO 5	Use decision trees for classification	3	2	2	1	2							1		2	2	
CO 6	Apply ensemble learning techniques for improved classification	3	2	2	1	2							1		2	2	

Note: CO to PO & PSO mapping level (1 – low, 2- moderate and 3 – substantial).